

Intermediate Macro

Chapter 22: Fiscal Policy (A Summing Up)

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22.1 What We Have Learned

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22.2 Ricardian Equivalence, Cyclical Adjusted Deficits, and

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22.5 The Challenges Facing US Fiscal Policy Today



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- The **inflation-adjusted deficit** is the deficit measured in real terms. It equals real interest payments plus the (real) primary deficit.
- The **primary deficit** is government spending minus taxes.
- The **cyclically-adjusted deficit** measures what the deficit would be if it were at its natural level.
- **Ricardian equivalence** is the proposition that neither deficits nor debts have any effect on economic activity.
- **Debt restructuring**, or **debt rescheduling** occurs when interest payments are deferred rather than cancelled.
- **Debt monetization** occurs when the central bank pays the government with money it creates, and the government uses that money to finance its deficit.
- **Hyperinflation** is a very high level of inflation.

22.1 What We Have Learned

- Navigation icons: back, forward, search, and other controls.

22.2 The Government Budget Constraint

- $$\text{deficit}_t = rB_{t-1} + G_t - T_t \quad (22.1)$$

- Note that we measure interest payment as *real* payments and the correct measure of the deficit is sometimes called the **inflation-adjusted deficit**.
- The **government budget constraint** then simply states that the *change in government debt during year t* is equal to the deficit during year t :

$$B_t - B_{t-1} = \text{deficit}_t$$

Government Budget Constraint

22.2 The Government Budget Constraint

- Using the definition of the deficit (equation (22.1)), we can rewrite the government budget constraint as

$$\underbrace{B_t - B_{t-1}}_{\text{Change in the Debt}} = \underbrace{rB_{t-1}}_{\text{Interest Payments}} + \underbrace{(G_t - T_t)}_{\text{Primary Deficit}} \quad (22.2)$$

- Moving B_{t-1} in the left side to the right side the equation and reorganizing,

$$B_t = (1 + r)B_{t-1} + (G_t - T_t) \quad (22.3)$$

Current versus Future Taxes

22.2 The Government Budget Constraint

- Consider first a one-year decrease in taxes for the path of debt and future taxes.
- Start from a situation where, until year 1, the government has balanced its budget, so that initial debt is equal to zero.
- During year 1, the government decreases taxes by 1 (think 1 billion dollars, for example) for one year.
- Thus, debt at the end of year 1 B_1 is equal to 1.
- The question we take up: What happens thereafter?

Full Repayment in Year 2

22.2 The Government Budget Constraint

- Suppose the government decides to fully repay the debt during year 2. From equation (22.3), the budget constraint for year 2 is given by

$$B_2 = (1 + r)B_1 + (G_2 - T_2)$$

- If the debt is fully repaid during year 2, then the debt at the end of year 2 is equal to zero, $B_2 = 0$.
- Replacing B_1 by 1 and B_2 by 0 and transposing terms gives

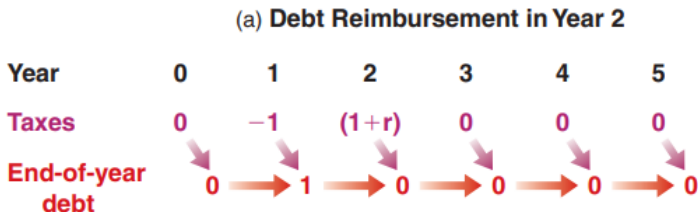
$$T_2 - G_2 = (1 + r)1 = (1 + r)$$

- To repay the debt fully during year 2, the government *must* run a primary surplus equal to $(1 + r)$.

Path of Taxes and Debt

22.2 The Government Budget Constraint

- The decrease in taxes by 1 during year 1 must be offset by an increase in taxes by $(1 + r)$ during year 2.



Full Repayment in Year t

22.2 The Government Budget Constraint

- Now suppose the government decides to wait until year t to repay the debt.
- From year 2 to year $t - 1$, the primary deficit is zero. So, from equation (22.3), debt at the end of year 2 is:

$$B_2 = (1 + r)B_1 + 0 = (1 + r)1 = (1 + r)$$

- With the primary deficit still equal to zero during year 3, debt at the end of year 3 is

$$B_3 = (1 + r)B_2 + 0 = (1 + r)(1 + r)1 = (1 + r)^2$$

- Note that debt grows at a rate equal to the interest rate, solving for debt at the end of year $t - 1$ is given by

$$B_{t-1} = (1 + r)^{t-2} \tag{22.4}$$

Full Repayment in Year t

22.2 The Government Budget Constraint

- In year t , the year in which the government decides to repay the debt, the budget constraint is

$$B_t = (1 + r)B_{t-1} + (G_t - T_t)$$

- If debt is fully repaid during year t , then B_t is zero. Replacing B_t by zero and B_{t-1} by its expression from equation (22.4) gives

$$0 = (1 + r)(1 + r)^{t-2} + (G_t - T_t)$$

- Reorganizing and bringing $(G_t - T_t)$ to the left side of the equation implies

$$T_t - G_t = (1 + r)^{t-1}$$

Debt Stabilization

22.2 The Government Budget Constraint

- Let us now look at what happens to taxes if the government stabilizes the debt (Stabilizing the debt means changing taxes or spending so that debt remains constant from then on).
- Suppose the government decides to stabilize the debt from year 2 on: the debt at the end of year 2 and thereafter remains at the same level as at the end of year 1.
- From equation (22.3), the budget constraint for year 2 is

$$B_2 = (1 + r)B_1 + (G_2 - T_2)$$

- Under our assumption that debt is stabilized in year 2, $B_2 = B_1 = 1$.
- Setting $B_2 = B_1 = 1$ in the preceding equation yields

$$1 = (1 + r)1 + (G_2 - T_2)$$

22.2 The Government Budget Constraint

- $$T_2 - G_2 = (1 + r) - 1 = r$$

Year	0	1	2	3	4	5
Taxes	0	-1	r	r	r	r
End-of-year debt	0	1	1	1	1	1

The Evolution of the Debt-to-GDP Ratio

22.2 The Government Budget Constraint

- We have focused so far on the evolution of the *level* of debt.
- But in an economy in which output grows over time, it makes more sense to focus instead on the *ratio of debt to output*.
- Divide both sides of equation (22.3) by real output Y to get

$$\frac{B_t}{Y_t} = (1 + r) \frac{B_{t-1}}{Y_t} + \frac{G_t - T_t}{Y_t}$$

- Next rewrite B_{t-1}/Y_t as $(B_{t-1}/Y_{t-1})(Y_{t-1}/Y_t)$:

$$\frac{B_t}{Y_t} = (1 + r) \left(\frac{Y_{t-1}}{Y_t} \right) \frac{B_{t-1}}{Y_{t-1}} + \frac{G_t - T_t}{Y_t}$$

The Evolution of the Debt-to-GDP Ratio

22.2 The Government Budget Constraint

- Assume that output growth is constant and denote the growth rate of output by g , so Y_{t-1}/Y_t can be written as $1/(1+g)$.
- And use the approximation $(1+r)/(1+g) \approx 1+r-g$.
- Using these two assumptions, rewrite the preceding equation as

$$\frac{B_t}{Y_t} = (1+r-g)\frac{B_{t-1}}{Y_{t-1}} + \frac{G_t - T_t}{Y_t}$$

- Finally, reorganize to get

$$\frac{B_t}{Y_t} - \frac{B_{t-1}}{Y_{t-1}} = (r-g)\frac{B_{t-1}}{Y_{t-1}} + \frac{G_t - T_t}{Y_t} \quad (22.5)$$

Ricardian Equivalence

22.3 Ricardian Equivalence, Cyclical Adjusted Deficits, and War Finance

- How does taking into account the government budget constraint affect the way we should think of the effects of deficits on output?
- One extreme view is that once the government budget constraint is taken into account, neither deficits nor debt have an effect on economic activity.
- This argument is known as the **Ricardian equivalence** proposition.
- The best way to understand the logic of the proposition is to use the example of tax changes from the previous section.
- Suppose that the government decreases by 1 (think 1 billion dollars) this year and announces that, to repay the debt, it will increase taxes by $(1 + r)$ next year. What will be the effect of the initial tax cut on consumption?

Logic of Ricardian Equivalence

22.3 Ricardian Equivalence, Cyclical Adjusted Deficits, and War Finance

- One possible answer is: No effect at all. Why? Because consumers realize that the tax cut is not much of a gift.
- Lower taxes this year are exactly offset, in present value, by higher taxes next yer.
- Another way of coming to the same answer—this time looking at saving rather than consumption—is as follows:
- To say that consumers do not change their consumption in response to the tax cut is the same as saying that *private saving increases one-for-one with the deficit*.
- In a nutshell, the decrease in public saving is offset by an equal increase in private saving.
- Total saving is therefore unaffected, and so is investment.
- How seriously should we take the Ricardian equivalence proposition?

Deficits, Output Stabilization, and the Cyclically Adjusted Deficit

22.3 Ricardian Equivalence, Cyclical Adjusted Deficits, and War Finance

- The fact that budget deficits do, indeed have long-run adverse effects on capital accumulation, and in turn adverse effects on output, does not imply that fiscal policy should not be used to reduce output fluctuations.
- Rather, it implies that deficits during recessions should be offset by surpluses during booms, so as not to lead to a steady increase in debt.
- The **cyclically-adjusted deficit** removes the effect of the business cycle from the deficit.
- Thus, it can be used to assess whether fiscal policy is consistent with no systematic increase in debt over time.

Construction of Cyclically Adjusted Deficit

22.3 Ricardian Equivalence, Cyclical Adjusted Deficits, and War Finance

- Estimating the cyclically-adjusted deficit requires knowledge of two facts:
 - ① the reduction in the deficit that would occur if output were to increase by 1% and
 - ② the difference between current output and its natural level.
- The first step is relatively straightforward. As a rule of thumb, a 1% decrease in output increases the deficit by 0.5% of GDP.
- The second step is more difficult. Too low an estimate of the natural rate of unemployment will lead too high an estimate of potential output and therefore too optimistic a measure of the cyclically adjusted deficit.

Wars and Deficits

22.3 Ricardian Equivalence, Cyclical Adjusted Deficits, and War Finance

- Wars typically bring about large budget deficits.
- Is it right for governments to rely so much on deficits to finance wars?
- There are two good reasons to run deficits during wars.
 - ① First, deficits shift part of the burden of paying for a war to future generations.
 - ② Second, by using debt instead of tax financing, the government avoids imposing very large and distortionary tax rates. Borrowing permits tax increases to be smoothed over time.

High Debt, Default Risk, and Vicious Cycles

22.4 The Dangers of High Debt

- We have seen that high debt requires high taxes in the future.
- A lesson from history is that high debt can also lead to vicious cycles, making the conduct of fiscal policy extremely difficult.
- Return to equation (22.5):

$$\frac{B_t}{Y_t} - \frac{B_{t-1}}{Y_{t-1}} = (r - g) \frac{B_{t-1}}{Y_{t-1}} + \frac{(G_t - T_t)}{Y_t}$$

- Take a country with a high initial debt ratio B_{t-1}/Y_{t-1} , say, 100%. Suppose the real interest rate is 3% and the growth rate is 2%.
- The first term on the right side is $(3\% - 2\%) \times 100\% = 1\%$ of GDP.
- Suppose further that the government is running a primary surplus of 1% of output, so just enough to keep the debt ratio constant (the right side of the equation equals 0%).

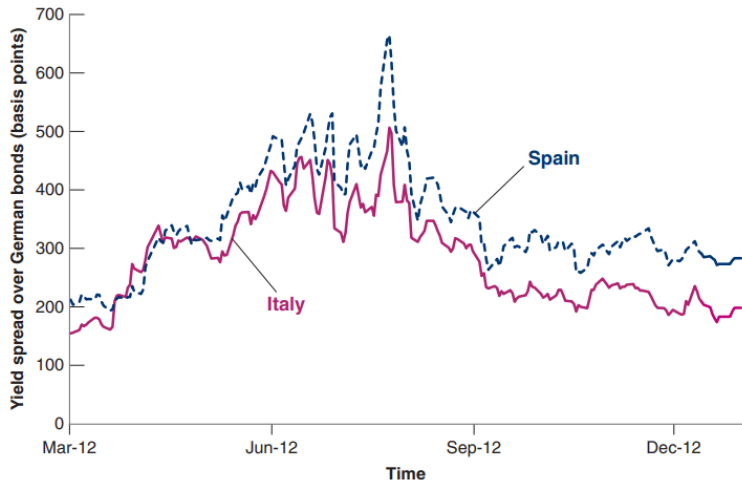
Effects of Increase in Interest Rate

22.4 The Dangers of High Debt

- Now suppose financial investors start to worry that the government may not be able to fully repay the debt.
- So, they ask for a higher interest rate to compensate for what they perceive as a higher risk of default on the debt.
- Suppose, for example, that the interest rate increases from 3% to 8%.
- Then, just to stabilize the debt, the government needs to run a primary surplus of 6% of output.
- In response to the increase in the interest rate, the government takes measures to increase the primary surplus to 6% of output.
- Increases in the interest rate and increases in the debt ratio feed on each other.

The Increase in European Bond Spreads

22.4 The Dangers of High Debt



Debt Default

22.4 The Dangers of High Debt

- If a government does not succeed in stabilizing the debt, historically, one of two things happens:
 - ① The government explicitly defaults on its debt or
 - ② The government relies increasingly on money finance, which typically leads to very high inflation.
- Default is often partial, and creditors take what is known as a **haircut**.
- Default also comes under many names, many of them euphemisms—probably to make the prospects more appealing (or less unappealing) to creditors.
- It is called **debt restructuring**, or **debt rescheduling** (when interest payments are deferred rather than cancelled), or, quite ironically, **private sector involvement**.

Money Finance

22.4 The Dangers of High Debt

- The government can finance itself by, in effect, printing money.
- The way it does this is not actually by printing money itself but by issuing bonds and then requiring the central bank to buy its bonds in exchange for money.
- This process is called **money finance** or **debt monetization**.
- Because, in this case, the rate of money creation is determined by the government deficit rather than by decisions of the central bank, this is also known as **fiscal dominance** of monetary policy.

Seignorage

22.4 The Dangers of High Debt

- How large a deficit can a government finance through such money creation?
- Let H be the amount of money of central bank money in the economy. Let ΔH be money creation.
- The revenue, in real terms, that the government generates by creating an amount of money equal to ΔH is therefore $\Delta H/P$.
- This revenue from money creation is called **seignorage**.

$$\text{seignorage} = \frac{\Delta H}{P}$$

- To see what rate of nominal money growth is required to generate a given amount of seignorage, rewrite $\Delta H/P$ as

$$\text{seignorage} = \frac{\Delta H}{P} = \frac{\Delta H}{H} \frac{H}{P}$$

Money Financing and Hyperinflation

22.4 The Dangers of High Debt

- To think about relevant magnitudes, it is convenient to take one more step and divide both sides of the equation by Y to get

$$\frac{\text{seignorage}}{Y} = \frac{\Delta H}{H} \left(\frac{H/P}{Y} \right) \quad (22.6)$$

- Suppose the government is running a budget deficit equal to 10% of GDP and decides to finance it through seignorage, so $(\text{deficit}/Y) = (\text{seignorage}/Y) = 10\%$.
- Choose $(H/P)/Y = 1$. This implies that nominal money growth must satisfy

$$10\% = \frac{\Delta H}{H} \times 1 \Rightarrow \frac{\Delta H}{H} = 10\%$$

- As money growth increases, inflation typically follows and high inflation turns into **hyperinflation**.

How Bad is the Debt Situation in the United States Today?

22.5 The Challenges Facing US Fiscal Policy Today

- Gross US debt in 2018 was approximately \$21,300 billion, or 104% of GDP. The most recent annual deficit was \$780 billion in that same year.
- When viewing the total debt as a percentage of the GDP, we note only a small increase in the ratio which gives little reason to worry.
- However, looking ahead there are some government spending items that will rapidly increase:
 - ① Social security payments are increasing rapidly and will continue to do so as more of the Baby Boom generation enters retirement.
 - ② Medicare and Medicaid are also projected to increase rapidly in the next several years for the same reason.

How Much and at What Rate Should Primary Deficits be Reduced?

22.5 The Challenges Facing US Fiscal Policy Today

- Given these future demands, it may be wise to begin efforts to lower the debt-to-GDP ratio.
- However, the current cost of debt service is very low given the historically low interest rates.
- Additionally, an argument can be made that an increase in public investment would be prudent because of the low interest rates.
- Perhaps a slow reduction in debt will work without pushing output down.

Summary

22.5 The Challenges Facing US Fiscal Policy Today

- The US debt-to-GDP ratio is high and slowly increasing over time.
- Given the likely increases in spending in the future, it would be desirable to reduce primary deficits and reduce the debt ratio.
- Low interest rates, and by implication the low cost of debt, imply that it can be done slowly.
- And because low interest rates, together with the zero lower bound, put sharp limits on the use of monetary policy, they also imply that it has to be done slowly so as to avoid a decrease in demand and output.